

Modeling the New AI Economy

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It is now apparent that Artificial Intelligence (AI) is having a major impact on the human economy. What we currently lack is a quantitative measure of those impacts and how government and society should adapt to the changes.

It is clear that AI will improve economic productivity while reducing the labor required for that productivity. What is not clear is whether new jobs will keep pace with displaced jobs. There are early indications that the pace of new job creation will NOT keep up with the job displacement. Despite positive economic headlines, there is a growing unease over the effects of AI on the labor economy.

The National Artificial Intelligence Economy Resilience Model (NAIERM) takes a systems engineering approach to modeling the dynamics of the emerging AI economy. Using Bureau of Economic Analysis metrics, a measure of the fraction of GDP produced from AI enhancement can be made. The model finds the fraction of GDP produced by the human component diminishes with time as the machine component increases.

As human labor income is displaced by machine capital, the rate at which income recirculates back into the economy begins to slow. Relatively quickly, there emerges a danger of insufficient demand to clear the supply.

By utilizing well-established metrics regarding the income recirculation rate to balance demand with supply, the model determines an optimum rate for a new, dedicated circulation path—the OASIS Circuit (Optimized Allocation for Sustained Income & Supply)—needed to maximize the GDP growth enabled by AI.

Crucially, the OASIS Circuit operates independently from normal government functions and the general federal budget. It is a sequestered, closed-loop system designed strictly to route AI-generated productivity directly to the consumer, immunizing it against political earmarks or ideologically-driven spending.

The optimized conditions of this circuit make every stakeholder—corporations, citizens, and government alike—beneficiaries of the improved productivity afforded by AI. Absent the OASIS Circuit, welfare requirements will continue to grow, the National debt will rise unsustainably, and the overall GDP will suffer. With the circuit in place, GDP grows, incomes appreciate, and the national debt can be retired by 2057.

While on the surface this new circulation path has similarities to the oft-discussed Universal Basic Income, it differs in both purpose and methodology. The governing function of the circulation rate tunes the relationship between demand and supply. Over time, as the economic engine revs, the volume of capital flowing through this circuit naturally expands, sustaining unprecedented GDP growth.

The cost to the tech sector to fund this consumer market is actually negative: the model shows that for every dollar of tax paid, the consumer market they sell into grows by more than three dollars.

The White paper describing the model and its results can be found at <https://ssrn.com/abstract=6858658>.